Communication lower bounds for Multi-Party Graph Traversal

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Abstract

A large number of computational problems can be modelled as communication games: For example, given a graph, two players each have a subgraph they know is traversable, and wish to collaboratively construct a path from one vertex to another. Communication complexity studies such situations by creating asymptotic bounds on the minimum communication necessary to complete these tasks. In this report, we prove communication lower and upper bounds for a variety of graph traversal problems, such as the example above. We also review potential applications of these bounds, such as for modelling real-world scenarios and designing algorithms.

Motivation

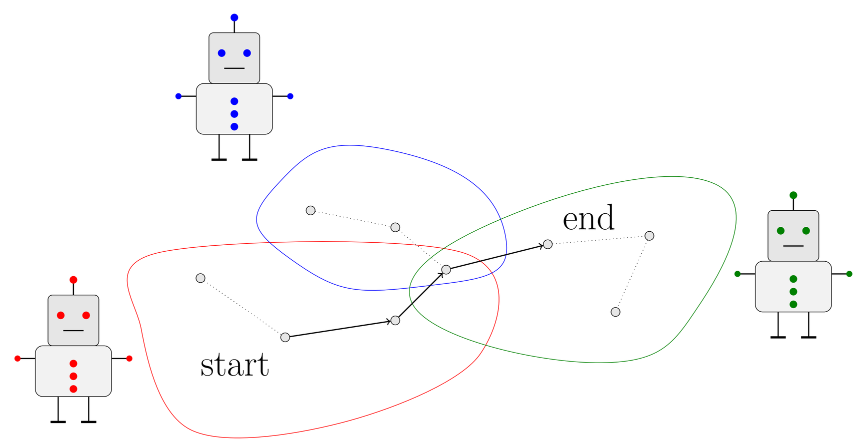

Figure 1:Path in Agent Owned Graphs

Communication complexity studies the fundamental lower bounds on how much information must be exchanged between agents to solve a distributed problem, no matter how clever the algorithm is. Thus by finding these lower bounds, we can design algorithms or learning functions that utilizes these lower bounds

In our problem, we consider a graph whose edges are split across agents. Each agent sees only its own edges. So thus given a start and end node how can they coordinate with each order in order to determine reachability.

Basic Example: Equality

There are two players: **Alice** and **Bob**.

- Alice holds input x
- Bob holds input y
- Together, they want to compute $f(x,y)$
- Goal: minimize the number of bits communicated

Equality $\mathrm{EQ}_n(x,y) = \begin{cases} 1 & \text{if } x = y, \\ 0 & \text{otherwise.} \end{cases}$

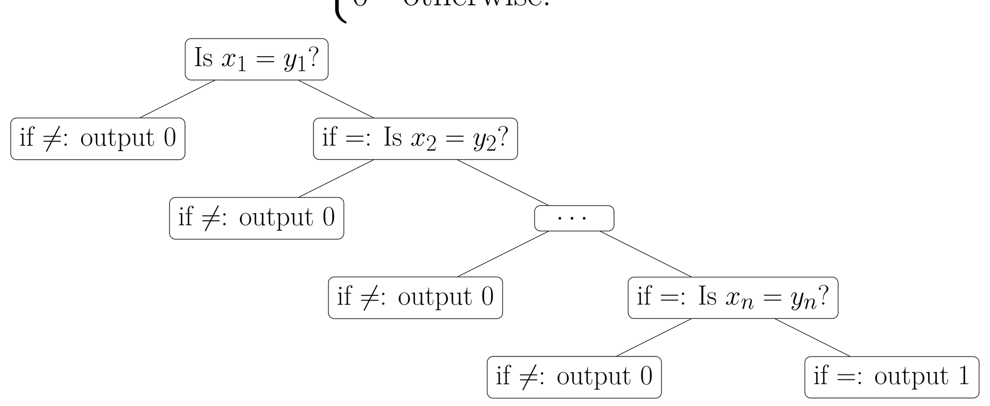

Figure 2:Visual for Equality Proof

Figure 3 demonstrates how, in the worst case, we must check every bit, since if every bit except x_n and y_n are the same, we will have to check all n bits to show equality. Thus, the deterministic communication complexity of Equality is $D(\mathrm{EQ}) = n$

Two Player TRAV with EQ

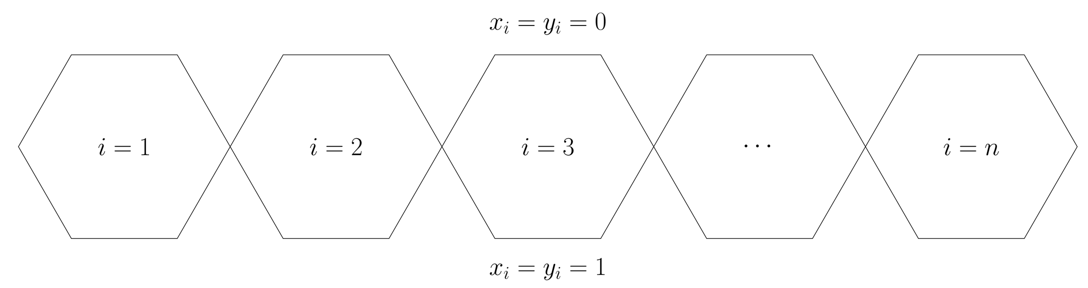

Figure 3:equality gadgets

1. Top branch corresponds to $x_i = y_i = 1$,
2. Bot branch corresponds to $x_i = y_i = 0$.

In this graph, there exist only a path from the first gadgets to last every gadgets are traversable, which occurs if and only if $x_i = y_i$ for all i, i.e. $x = y$

Multi-Player TRAV, disjoint graphs

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

$$E = mc^2 \tag{1}$$

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

$$\cos^3\theta = \frac{1}{4}\cos\theta + \frac{3}{4}\cos 3\theta \tag{2}$$

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

$$\kappa = \frac{\xi}{E_{\max}} \tag{3}$$

Two Player TRAV with DISJ.

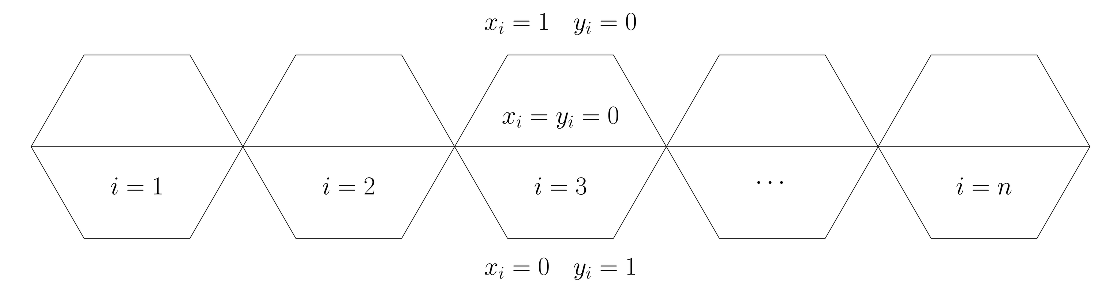

Figure 4:disjoint gadgets

1. Top branch corresponds to x_i has the element but not y_i
2. Mid branch corresponds to x_i and y_i do not have the element
3. Bot branch corresponds to y_i has the element but not x_i

In this graph, there exist only a path from the first gadgets to last every gadgets are traversable, which occurs if and only if $x \cap y_i = \varnothing$

Multi-Player TRAV, connected graphs

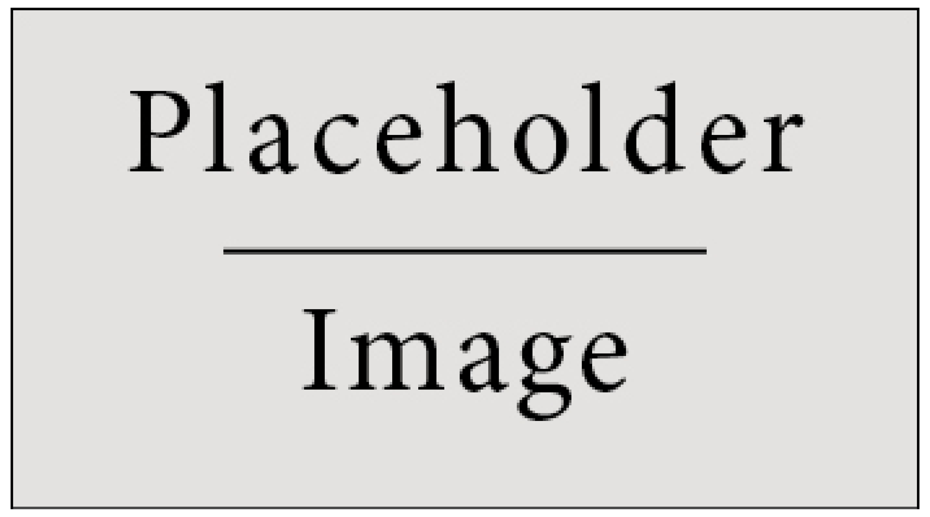

Figure 5:Figure caption

Nunc tempus venenatis facilisis. Curabitur suscipit consequat eros non porttitor. Sed a massa dolor, id ornare enim:

Treatments	Response 1	Response 2
Treatment 1	0.0003262	0.562
Treatment 2	0.0015681	0.910
Treatment 3	0.0009271	0.296

Table 1:Table caption

Two-player TRAV, iterative

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Multi-Player TRAV, iterative

Maecenas ultricies feugiat velit non mattis. Fusce tempus arcu id ligula varius dictum.

- Curabitur pellentesque dignissim
- Eu facilisis est tempus quis
- Duis porta consequat lorem

References

Acknowledgements

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